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Semiconductor module

Abstract

The present disclosure includes: a base plate having a shape of a sheet; a relay plate having a shape of a sheet; a terminal member; and an electronic component joined to one surface of the base plate. The base plate, the relay plate, and the terminal member are electrically conductive members and arranged on a same plane with gaps between the electrically conductive members. The electronic component and one surface of the relay plate are connected to each other by a bonding wire. The one surface of the relay plate and one surface of the terminal member are connected to each other by a bonding wire.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/JP2020/019718 filed May 19, 2020.

TECHNICAL FIELD

(2) The present disclosure relates to a semiconductor module.

BACKGROUND ART

(3) A semiconductor module including therein a semiconductor switching element is used for a power conversion device or the like. The semiconductor module is obtained by mounting a plurality of semiconductor switching elements on a base plate serving as an electrically conductive member, connecting, by means of jumper wires or bonding wires, electrode pads provided on the upper surfaces of the semiconductor switching elements, and then sealing these components with a mold resin. A plurality of terminal portions such as positive and negative power supply terminals, an output terminal for a load, and a control terminal for controlling each semiconductor switching element are arrayed on side surfaces on the outer periphery of the mold resin. In addition to the plurality of semiconductor switching elements, a shunt resistor for current detection and the like are also mounted inside the semiconductor module.

(4) If the length of a bonding wire provided inside the semiconductor module is excessively large, wire sagging, or wire sweep due to flow of the mold resin in a mold resin molding step, might occur. The occurrence of the wire sagging or the wire sweep leads to a problem that wire disconnection or short-circuiting due to contact between wires might occur. A three-dimensional wiring structure in which a relay wire member is disposed on a base plate, and a structure in which a relay for connection is performed via the relay wire member by means of wire bonding performed at a plurality of positions, have been disclosed to inhibit wire sagging and wire sweep (see, for example, Patent Document 1).

CITATION LIST

Patent Document

(5) Patent Document 1: Japanese Laid-Open Patent Publication No. 2018-125424

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(6) In an internal structure of a semiconductor module in the above Patent Document 1, bonding wires are connected via the relay wire member, in the three-dimensional wiring structure, which is disposed on the base plate. Consequently, wire sagging and wire sweep can be inhibited. However, since the wiring structure is three-dimensional, the height of a loop of the bonding wires is increased. Thus, the thickness of the mold resin is increased, resulting in a problem that the semiconductor module is upsized.

(7) Considering this, an object of the present disclosure is to provide a semiconductor module in which bonding wire sagging and bonding wire sweep are inhibited, and furthermore, the thickness

of a mold resin which performs sealing for the semiconductor module is reduced.

Solution to the Problems

(8) A semiconductor module according to the present disclosure includes: a base plate having a shape of a sheet; a relay plate having a shape of a sheet; a terminal member; and an electronic component joined to one surface of the base plate. The base plate, the relay plate, and the terminal member are electrically conductive members and arranged on a same plane with gaps between the electrically conductive members. The electronic component and one surface of the relay plate are connected to each other by a bonding wire. The one surface of the relay plate and one surface of the terminal member are connected to each other by a bonding wire.

Effect of the Invention

(9) In the semiconductor module according to the present disclosure, the base plate, the relay plate, and the terminal member are electrically conductive members and arranged on the same plane with gaps between the electrically conductive members, the electronic component and the one surface of the relay plate are connected to each other by a bonding wire, and the one surface of the relay plate and the one surface of the terminal member are connected to each other by a bonding wire. Consequently, bonding wire sagging and bonding wire sweep can be inhibited, and furthermore, the thickness of a mold resin which performs sealing for the semiconductor module can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a circuit configuration of a device including a semiconductor module according to embodiment 1.

(2) FIG. 2 is a plan view showing the semiconductor module according to embodiment 1.

(3) FIG. 3 is a side view showing the semiconductor module according to embodiment 1.

(4) FIG. 4 is a plan view showing a semiconductor module according to embodiment 2.

(5) FIG. 5A is a diagram for explaining a semiconductor module in a comparative example.

(6) FIG. 5B is a diagram for explaining a semiconductor module in a comparative example.

DESCRIPTION OF EMBODIMENTS

(7) Hereinafter, semiconductor modules according to embodiments of the present disclosure will be described with reference to the drawings. Description will be given while the same or corresponding members and parts in the drawings are denoted by the same reference characters.

Embodiment 1

(8) FIG. 1 illustrates a circuit configuration of the entirety of a system of an electric power steering device **200** including a semiconductor module **100** according to embodiment 1. FIG. 2 is a plan view showing an internal configuration of the semiconductor module **100**. FIG. 3 is a side view showing a major part of the internal configuration of the semiconductor module **100**. In each of FIG. 2 and FIG. 3, a mold resin **70** is excluded, and the alternate long and two short dashes line indicates the external shape of the mold resin **70**. FIG. 3 is a side view as seen in the direction of the arrow A shown in FIG. 2. The semiconductor module **100** is a module that is used for a power conversion device or the like and that includes therein a plurality of semiconductor switching elements **41**, **42**, and **44** for performing power conversion.

(9) <Electric Power Steering Device **200**>

(10) The electric power steering device **200** will be described as an example of the device including the semiconductor module **100**. The electric power steering device **200** is a device that converts power supplied from a battery **9** and that outputs power obtained by the conversion to a motor **2**. As shown in FIG. 1, the electric power steering device **200** is composed of a control unit **1**, the motor **2**, the battery **9**, an ignition switch **10**, and sensors such as a rotation sensor **6**. The motor **2** is a brushless motor composed of windings for three phases (a U phase, a V phase, and a W phase). The

rotation sensor **6** is disposed near the motor **2**, detects a rotation angle of the motor **2**, and outputs the rotation angle to a control circuit **3** of the control unit **1**. A sensor **8** includes a torque sensor for detecting steering wheel operation force, a vehicle speed sensor, and the like, and outputs an electric signal based on physical quantities detected by the respective sensors to the control circuit **3**.

(11) The control unit **1** is composed of: the control circuit **3** including a CPU **30**; an inverter circuit **4** for supplying current to the windings of the motor **2**; a power supply relay **5**; and other electrical components. The control unit **1** is connected to the battery **9** and the ignition switch **10**, and power is supplied to the control unit **1** via the battery **9** and the ignition switch **10**. The control unit **1** is connected to the rotation sensor **6** and the sensor **8**, and electric signals outputted from these sensors are inputted to the control circuit **3** of the control unit **1**.

(12) The control circuit **3** is composed of: the CPU **30**; a drive circuit **31**; an input circuit **32**; and a power supply circuit **33** connected to these circuits. Output signals from the rotation sensor **6** and the sensor **8** are inputted to the input circuit **32**, and data, obtained from these output signals, such as rotation angle information about the motor **2** are outputted to the CPU **30**. The CPU **30** calculates, from the inputted data, a current value for rotating the motor **2**. The CPU **30** outputs a control signal corresponding to the result obtained by the calculation, to the inverter circuit **4** via the drive circuit **31**. In addition, a current detection signal detected by the inverter circuit **4** is inputted to the CPU **30** via the input circuit **32**, and the CPU **30** performs feedback control on the inverter circuit **4** on the basis of the current detection signal.

(13) The inverter circuit **4** is composed of bridge circuits corresponding to respective windings for the three phases, i.e., the U phase, the V phase, and the W phase, of the motor **2**. All of the bridge circuits corresponding to the respective phases are identical to one another, and thus a bridge circuit **4a** for one phase (U phase) will be described. The bridge circuit **4a** includes three semiconductor switching elements **41**, **42**, and **44**. The semiconductor switching element **41** on a high-potential side and the semiconductor switching element **42** on a low-potential side are connected in series. One side of the semiconductor switching element **44** is connected to an intermediate connection portion between the semiconductor switching element **41** and the semiconductor switching element **42**, and another side of the semiconductor switching element **44** is connected to a U-phase coil of the motor **2**. The semiconductor switching element **44** has a relay function of enabling connection or disconnection between the U-phase coil and the inverter circuit **4**.

(14) A shunt resistor **43** for detecting current is connected downstream of the semiconductor switching element **42** on the low-potential side. A potential difference between both ends of the shunt resistor **43** is detected and converted into a current value. In the inverter circuit **4**, terminals **48** are provided at both ends of the shunt resistor **43** in order to detect the potential difference between both ends of the shunt resistor **43**. Monitor signals at various points of the inverter circuit **4** are inputted to the CPU **30** via the input circuit **32**. Drive of the semiconductor switching elements **41**, **42**, and **44** is controlled on the basis of a signal outputted from the drive circuit **31** according to a command from the CPU **30**.

(15) The power supply relay **5** is disposed upstream of the inverter circuit **4**. The power supply relay **5** includes two semiconductor switching elements **5a** and **5b** connected in series. The power supply relay **5** has a relay function of enabling supply or interruption of power to the inverter circuit **4**. Capacitors **7** for smoothing are connected in parallel between a ground and a power supply supplied to the inverter circuit **4**.

(16) In the case where the control unit **1** is configured by using a semiconductor module including therein a plurality of semiconductor switching elements, various types of semiconductor modules are conceived. Examples of the semiconductor modules include: a semiconductor module including therein a bridge circuit composing the inverter circuit **4**; a semiconductor module including therein the power supply relay **5**; and a semiconductor module including therein a circuit obtained by eliminating some of the shunt resistors and the semiconductor switching elements.

(17) <Semiconductor Module **100**>

(18) A case where the bridge circuit **4a** for the one phase is formed as a semiconductor module **100** will be described. As shown in FIG. 2, the semiconductor module **100** includes: base plates **50** each having the shape of a sheet; a relay plate **60** having the shape of a sheet; terminal members **80**; and electronic components each joined to one surface of a corresponding one(s) of the base plates **50**. The base plates **50**, the relay plate **60**, and the terminal members **80** are electrically conductive members made from a same material such as copper or a copper alloy. The base plates **50**, the relay plate **60**, and the terminal members **80** are provided side by side on a same plane with gaps between the electrically conductive members. Here, five base plates **50** and four terminal members **80** are provided. The electronic components of the semiconductor module **100** are the semiconductor switching elements **41**, **42**, and **44** and the shunt resistor **43**. The semiconductor switching elements **41**, **42**, and **44** described herein are field effect transistors (FETs). It is noted that the electronic components joined to the base plates **50** are not limited thereto.

(19) The semiconductor switching element **41** on the high-potential side is, at a drain portion of the semiconductor switching element **41**, joined to one surface of a base plate **50a**. The semiconductor switching element **42** on the low-potential side is, at a drain portion of the semiconductor switching element **42**, joined to one surface of a base plate **50c**. The semiconductor switching element **44** is, at a drain portion of the semiconductor switching element **44**, joined to one surface of a base plate **50b**. The shunt resistor **43** is joined on and between one surface of a base plate **50d** and one surface of a base plate **50e**. The semiconductor switching elements **41**, **42**, and **44** and base plates **50** are connected by, for example, jumper wires **45** each formed of a copper sheet and each having the shape of a bridge. A jumper wire **45a** connects a source portion of the semiconductor switching element **41**, a source portion of the semiconductor switching element **44**, and the base plate **50c** to each other. A jumper wire **45b** connects a source portion of the semiconductor switching element **42** and the base plate **50d** to each other.

(20) The base plates **50** include terminal portions **51** exposed to outside from side surfaces of the mold resin **70**. Directions in which the terminal portions **51** are exposed are directions parallel to the same plane on which the base plates **50** are provided. A terminal portion **51a** of the base plate **50a** is connected to the battery **9** which is a power supply. A terminal portion **51e** of the base plate **50e** is connected to the ground. A terminal portion **51b** of the base plate **50b** is connected to the U phase of the motor **2** which is a load, and power obtained through conversion is outputted from the terminal portion **51b** to the U phase of the motor **2**. The terminal portions **51a**, **51b**, and **51e** are high-current power terminals having large current amounts. In FIG. 2, the terminal portion **51a** and the terminal portion **51e** are disposed side by side on the right side of the semiconductor module **100**. In FIG. 2, the terminal portion **51b** is provided to extend in the upward direction from the semiconductor module **100**.

(21) The base plate **50d** and the base plate **50e** respectively include a terminal portion **51d** and a terminal portion **51e1** for detecting a voltage between both ends of the shunt resistor **43**. The terminal portion **51e** and the terminal portion **51e1** are each at the ground potential. The base plate **50c** includes a terminal portion **51c** connected to a connection point between the semiconductor switching elements **41**, **42**, and **44**. The terminal portion **51c** is a terminal for monitoring the potential of the connection point between the semiconductor switching elements **41**, **42**, and **44**.

(22) Each terminal member **80** having the shape of a sheet includes a terminal portion **81** exposed to outside from a side surface of the mold resin **70**. A direction in which the terminal portion **81** is exposed is a direction parallel to a same plane on which the terminal members **80** are provided. Each terminal member **80** is a terminal that has a relatively small current amount and that is used for controlling the semiconductor module **100**. In FIG. 2, the terminal members **80** are disposed side by side on the lower side of the semiconductor module **100**. Terminal members **80a**, **80b**, and **80c** are gate control terminals connected to gate portions of the respective semiconductor switching elements **41**, **42**, and **44**. Each of the terminal members **80a**, **80b**, and **80c** and the corresponding

gate portion are connected to each other by a bonding wire **46** provided in a wire bonding step. Each of the terminal members **80b** and **80c** and the corresponding gate portion are directly connected to each other by a bonding wire **46**. Regarding the terminal member **80a** and the gate portion of the semiconductor switching element **41**, the gate portion and one surface of the relay plate **60** are connected to each other by a bonding wire **46a**, and the one surface of the relay plate **60** and one surface of the terminal member **80a** are connected to each other by a bonding wire **46b**. By connecting the terminal member **80a** and the gate portion to each other with division into the bonding wire **46a** and the bonding wire **46b**, the length of each bonding wire is shortened as compared to the case where the terminal member **80a** and the gate portion are directly connected to each other by a single bonding wire. Consequently, bonding wire sagging and bonding wire sweep can be inhibited.

(23) Wire bonding is performed from a start point which is the gate portion of each of the semiconductor switching elements **41**, **42**, and **44** to an end point which is a point on the corresponding terminal member **80** side. Thus, the bonding wires **46** can be cut without damaging the semiconductor switching elements **41**, **42**, and **44** in a wire bonding step. Although the gate portion of the semiconductor switching element **41** and the terminal member **80a** are connected to each other via the relay plate **60**, the start point and the end point are unchanged. A terminal member **80d** and the base plate **50b** are connected to each other by a bonding wire **47**. The terminal member **80d** is a terminal for monitoring the potential of the terminal portion **51b** from which output to the motor **2** is performed.

(24) The mold resin **70** seals: the base plates **50**; the relay plate **60**; the terminal members **80**; the semiconductor switching elements **41**, **42**, and **44** and the shunt resistor **43** which are electronic components; and the bonding wires **46** and **47**. The mold resin **70** is formed in an external shape that is the shape of a rectangular sheet having a sheet surface parallel to the same plane on which the base plates **50** are provided. Another surface of each base plate **50** may be exposed from the mold resin **70** in order to easily cool the base plate **50**.

(25) <Comparative Example>

(26) A comparative example will be described with reference to FIG. 5 before the relay plate **60** which is a major part of the present disclosure is described. FIG. 5 is a diagram for explaining a semiconductor module **101** in the comparative example. FIG. 5A is a plan view showing a major part of the internal configuration of the semiconductor module **101**. FIG. 5B is a side view showing the major part of the internal configuration of the semiconductor module **101**. The base plate **50b** and a relay plate **60a** are connected to each other by a jumper wire **45c** which is formed of a copper sheet and which has the shape of a bridge. The gate portion of the semiconductor switching element **41** and the terminal member **80a** are connected to each other by a bonding wire **46c** provided in a wire bonding step. Wire bonding between the gate portion and the terminal member **80a** cannot be performed such that the bonding wire **46c** extends over the jumper wire **45c**. Thus, the bonding wire **46c** is provided via, as a relay point, the jumper wire **45c** on the relay plate **60a**. The jumper wire **45c** is at a higher position than one surface of each of the base plate **50a** and the relay plate **60a**, and thus the height of a loop of the bonding wire **46c** increases at the positions indicated by the arrows in FIG. 5(b). Since the height of the loop of the bonding wire **46c** increases, the thickness of the mold resin **70** increases, whereby the semiconductor module **101** is upsized.

(27) <Relay Plate **60**>

(28) The relay plate **60** which is a major part of the present disclosure will be described. As shown in FIG. 2, the relay plate **60** has the shape of a rectangular sheet. If a direction parallel to a long side of the relay plate **60** is defined as a long side direction, each bonding wire **46** is connected to a one-side portion in the long side direction of the relay plate **60**. If the relay plate **60** has the shape of a sheet, the relay plate **60** can be made from a material that is the same as the materials of the base plates **50** and can be disposed side by side with the base plates **50** on a plane that is the same as the plane of the base plates **50**. Since the relay plate **60** can be disposed side by side with the

base plates **50** on the plane that is the same as the plane of the base plates **50**, the height of the loop of the bonding wire **46** connected to the relay plate **60** can be made low as shown in FIG. **3**. Since the height of the loop of the bonding wire **46** can be reduced, the thickness of the mold resin **70** is reduced, whereby the semiconductor module **100** can be downsized. If the relay plate **60** has a rectangular shape, the size thereof can be set such that the relay plate **60** only has an area necessary for wire bonding work. Therefore, the semiconductor module **100** can be downsized.

(29) Although the bonding wire **46** is connected to the one-side portion in the long side direction herein, the bonding wire **46** may be connected to an other-side portion in the long side direction. If the bonding wire **46** is connected to the one-side portion or the other-side portion in the long side direction of the relay plate **60**, the other-side portion or the one-side portion to which the bonding wire **46** is not connected can be held down by using equipment such as a jig at the time of wire bonding work so as to fix the relay plate **60**. Since the relay plate **60** can be easily fixed at the time of wire bonding work, productivity for the semiconductor module **100** can be improved.

(30) The semiconductor switching element **41** which is an electronic component is disposed on one side in the long side direction relative to the relay plate **60**, the terminal member **80a** is disposed on another side in the long side direction relative to the relay plate **60**, and the bonding wire **46** is connected to the one-side portion in the long side direction of the relay plate **60**. In the case of this arrangement, wire bonding work can be performed in the long side direction, and thus the productivity for the semiconductor module **100** can be improved. In addition, the length of the bonding wire **46** can be shortened. In addition, the semiconductor switching element **41** can be set as a start point for wire bonding, and the terminal member **80a** can be set as an end point for the wire bonding, whereby the bonding wire **46** can be cut without damaging the semiconductor switching element **41**.

(31) The bonding wire **46a** connecting the semiconductor switching element **41** and the one surface of the relay plate **60** to each other, and the bonding wire **46b** connecting the one surface of the relay plate **60** and the one surface of the terminal member **80a** to each other, constitute a single bonding wire **46**. In the case where the bonding wire **46a** and the bonding wire **46b** constitute a single bonding wire **46**, the semiconductor switching element **41** and the terminal member **80a** can be connected to each other through continuous-wire bonding work without cutting the bonding wire **46** at an intermediate portion thereof. Consequently, the productivity for the semiconductor module **100** can be improved.

(32) The relay plate **60** is disposed such that one long side thereof extends along a side surface of the mold resin **70**. In the case where the relay plate **60** is disposed along the side surface of the mold resin **70** instead of being disposed at a position enclosed by the base plates **50**, a gap portion generated between the relay plate **60** and each base plate **50** can be reduced, whereby the semiconductor module **100** can be downsized. In addition, the degree of freedom in the layout of the base plates **50** can be improved. In addition, even though the relay plate **60** is provided, the size of each base plate **50** is ensured in the same manner as in a conventional configuration, and thus the heat dissipation property of the semiconductor module **100** can be maintained.

(33) As described above, in the semiconductor module **100** according to embodiment **1**, the base plates **50**, the relay plate **60**, and the terminal members **80** are electrically conductive members and arranged on a same plane with gaps between the electrically conductive members, the semiconductor switching element **41** and the one surface of the relay plate **60** are connected to each other by the bonding wire **46a**, and the one surface of the relay plate **60** and the one surface of the terminal member **80a** are connected to each other by the bonding wire **46b**. Consequently, bonding wire sagging and bonding wire sweep are inhibited, and furthermore, the thickness of the mold resin **70** which performs sealing for the semiconductor module **100** can be reduced. In the case where the relay plate **60** has a rectangular shape, the size thereof can be set such that the relay plate **60** only has an area necessary for wire bonding work. Therefore, the semiconductor module **100** can be downsized.

(34) In the case where the bonding wire **46** is connected to the one-side portion or the other-side portion in the long side direction of the relay plate **60**, the other-side portion or the one-side portion to which the bonding wire **46** is not connected can be held down by using equipment such as a jig at the time of wire bonding work so as to easily fix the relay plate **60**. Therefore, the productivity for the semiconductor module **100** can be improved. In the case where the semiconductor switching element **41** is disposed on the one side in the long side direction relative to the relay plate **60**, the terminal member **80a** is disposed on the other side in the long side direction relative to the relay plate **60**, and the bonding wire **46** is connected to the one-side portion in the long side direction of the relay plate **60**, wire bonding work can be performed in the long side direction, whereby the productivity for the semiconductor module **100** can be improved. In addition, the length of the bonding wire **46** can be shortened. In addition, the semiconductor switching element **41** can be set as a start point for wire bonding, and the terminal member **80a** can be set as an end point for the wire bonding, whereby the bonding wire **46** can be cut without damaging the semiconductor switching element **41**.

(35) In the case where the bonding wire **46a** connecting the semiconductor switching element **41** and the one surface of the relay plate **60** to each other, and the bonding wire **46b** connecting the one surface of the relay plate **60** and the one surface of the terminal member **80a** to each other, constitute a single bonding wire **46**, the semiconductor switching element **41** and the terminal member **80a** can be connected to each other through continuous-wire bonding work without cutting the bonding wire **46** at an intermediate portion thereof. Consequently, the productivity for the semiconductor module **100** can be improved. In the case where the relay plate **60** is disposed such that one long side thereof extends along a side surface of the mold resin **70**, a gap portion generated between the relay plate **60** and each base plate **50** can be reduced, whereby the semiconductor module **100** can be downsized. In addition, the degree of freedom in the layout of the base plates **50** can be improved. In addition, even though the relay plate **60** is provided, the size of each base plate **50** is ensured in the same manner as in a conventional configuration, and thus the heat dissipation property of the semiconductor module **100** can be maintained.

(36) In the case where the base plates **50**, the relay plate **60**, and the terminal members **80** are made from a same material, the base plates **50**, the relay plate **60**, and the terminal members **80** can be made from a single sheet material without adding any separate member. Consequently, the productivity for the semiconductor module **100** can be improved.

Embodiment 2

(37) A semiconductor module **100** according to embodiment 2 will be described. FIG. 4 is a plan view showing an internal configuration of the semiconductor module **100** according to embodiment 2. In FIG. 4, the mold resin **70** is excluded, and the alternate long and two short dashes line indicates the external shape of the mold resin **70**. The semiconductor module **100** according to embodiment 2 includes, in addition to the components described in embodiment 1, extending-out portions **52** and **61** exposed to outside from side surfaces of the mold resin **70**.

(38) Each of the base plates **50**, the relay plate **60**, and the terminal members **80** includes a terminal portion **51**, **81** or an extending-out portion **52**, **61** exposed to outside from a side surface of the mold resin **70**. Directions in which the terminal portions **51** and **81** and the extending-out portions **52** and **61** are exposed are directions parallel to the same plane on which the base plates **50** are provided. The terminal portions **51** and **81** and the extending-out portions **52** and **61** are connected to an outer frame **90** indicated by the broken lines in FIG. 4 at the time of manufacturing the semiconductor module **100**. The base plates **50**, the relay plate **60**, the terminal members **80**, and the outer frame **90** are formed of a single sheet material made from copper, a copper alloy, or the like.

(39) The semiconductor module **100** is manufactured in a state where the base plates **50**, the relay plate **60**, and the terminal members **80** are connected to the outer frame **90**. A method for manufacturing the semiconductor module **100** includes: a step of punching an electrically

conductive member having the shape of a sheet, to form a member in which the base plates **50**, the relay plate **60**, and the terminal members **80** are connected to the outer frame **90**; a step of joining, to the base plate **50**, the semiconductor switching elements **41**, **42**, and **44** and the shunt resistor **43** which are electronic components; a step of connecting the semiconductor switching elements **41**, **42**, and **44** and the base plates **50** by the jumper wires **45**; a step of connecting the semiconductor switching elements **41**, **42**, and **44**, the relay plate **60**, and the terminal members **80** by the bonding wires **46** and **47**; a step of sealing the base plates **50**, the relay plate **60**, the terminal members **80**, the semiconductor switching elements **41**, **42**, and **44**, the shunt resistor **43**, and the bonding wires **46** and **47** by the mold resin **70**; and a step of cutting off the outer frame **90** from the terminal portions **51** and **81** and the extending-out portions **52** and **61**.

(40) By cutting off the outer frame **90** from the terminal portions **51** and **81** and the extending-out portions **52** and **61**, the semiconductor module **100** is completed. Since each of the base plates **50**, the relay plate **60**, and the terminal members **80** includes a terminal portion **51**, **81** or an extending-out portion **52**, **61** exposed to outside from a side surface of the mold resin **70**, the arrangement of the base plates **50**, the relay plate **60**, and the terminal members **80** can be stabilized at the time of manufacturing the semiconductor module **100**, without using another fixation member such as a jig. Since the arrangement of the base plates **50**, the relay plate **60**, and the terminal members **80** is stabilized, the productivity for the semiconductor module **100** can be improved.

(41) The extending-out portion **61** of the relay plate **60** will be described. The width of the extending-out portion **61** of the relay plate **60** is shorter than the length of the long side of the relay plate **60**. In the case where the width of the extending-out portion **61** is set to be shorter than the length of the long side of the relay plate **60**, the extending-out portion **61** can be easily cut off from the outer frame **90**. Since the extending-out portion **61** can be easily cut off from the outer frame **90**, the productivity for the semiconductor module **100** can be improved.

(42) The extending-out portion **61** of the relay plate **60** is provided on a side, on which there is the one long side of the relay plate **60**, so as to be located at the other-side portion in the long side direction of the relay plate **60**. In FIG. 4, the extending-out portion **61** is provided at the other-side portion, of the relay plate **60**, to which the bonding wire **46** is not connected. In the case where the extending-out portion **61** is provided on the side on which the bonding wire **46** is not connected, the area of the relay plate **60** at a portion thereof on the side on which the bonding wire **46** is not connected is enlarged. Consequently, this portion can be easily held down by using equipment such as a jig at the time of wire bonding work so as to fix the relay plate **60**. Since the relay plate **60** can be easily fixed at the time of wire bonding work, the productivity for the semiconductor module **100** can be improved. In addition, the relay plate **60**, inclusive of the extending-out portion **61**, can be held down by using equipment such as a jig. Consequently, the area of the relay plate **60** can be reduced, and the relay plate **60** can be downsized. Since the relay plate **60** can be downsized, the semiconductor module **100** can be downsized.

(43) As described above, in the semiconductor module **100** according to embodiment 2, each of the base plates **50**, the relay plate **60**, and the terminal members **80** includes a terminal portion **51**, **81** or an extending-out portion **52**, **61** exposed to outside from a side surface of the mold resin **70**, and the directions in which the terminal portions **51** and **81** and the extending-out portions **52** and **61** are exposed are directions parallel to the same plane on which the base plates **50** are provided. Consequently, at the time of manufacturing, the terminal portions **51** and **81** and the extending-out portions **52** and **61** can be easily connected to the outer frame **90** which is to be cut off therefrom. Therefore, the arrangement of the base plates **50**, the relay plate **60**, and the terminal members **80** at the time of manufacturing the semiconductor module **100** can be stabilized. In addition, since the arrangement of the base plates **50**, the relay plate **60**, and the terminal members **80** is stabilized, the productivity for the semiconductor module **100** can be improved.

(44) In the case where the width of the extending-out portion **61** is set to be shorter than the length of the long side of the relay plate **60**, the extending-out portion **61** can be easily cut off from the

outer frame **90**. Consequently, the productivity for the semiconductor module **100** can be improved. In the case where the extending-out portion **61** of the relay plate **60** is provided on a side, on which there is one long side of the relay plate **60**, so as to be located at the one-side portion or the other-side portion, in the long side direction of the relay plate **60**, which is on the side on which the bonding wire **46** is not connected, the area at the portion on the side on which the bonding wire **46** is not connected is enlarged. Consequently, this portion can be easily held down by using equipment such as a jig at the time of wire bonding work so as to fix the relay plate **60**. Since the relay plate **60** can be easily fixed at the time of wire bonding work, the productivity for the semiconductor module **100** can be improved. In addition, since the relay plate **60**, inclusive of the extending-out portion **61**, can be held down by using equipment such as a jig, the relay plate **60** can be downsized. Since the relay plate **60** can be downsized, the semiconductor module **100** can be downsized.

(45) Although the disclosure is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features, aspects, and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead can be applied, alone or in various combinations to one or more of the embodiments of the disclosure.

(46) It is therefore understood that numerous modifications which have not been exemplified can be devised without departing from the scope of the specification of the present disclosure. For example, at least one of the constituent components may be modified, added, or eliminated. At least one of the constituent components mentioned in at least one of the preferred embodiments may be selected and combined with the constituent components mentioned in another preferred embodiment.

DESCRIPTION OF THE REFERENCE CHARACTERS

(47) **1** control unit **2** motor **3** control circuit **4** inverter circuit **4a** bridge circuit **5** power supply relay **5a** semiconductor switching element **5b** semiconductor switching element **6** rotation sensor **7** capacitor **8** sensor **9** battery **10** ignition switch **30** CPU **31** drive circuit **32** input circuit **33** power supply circuit **41** semiconductor switching element **42** semiconductor switching element **43** shunt resistor **44** semiconductor switching element **45** jumper wire **46** bonding wire **47** bonding wire **48** terminal **50** base plate **51** terminal portion **52** extending-out portion **60** relay plate **60a** relay plate **61** extending-out portion **70** mold resin **80** terminal member **81** terminal portion **90** outer frame **100** semiconductor module **101** semiconductor module **200** electric power steering device

Claims

1. A semiconductor module comprising: a base plate having a shape of a sheet; a relay plate having a shape of a sheet; a terminal member; an electronic component joined to one surface of the base plate; and a mold resin sealing the base plate, the relay plate, the terminal member, the electronic component, and bonding wire, wherein the base plate, the relay plate, and the terminal member are electrically conductive members and arranged as a same plane with gaps between the electrically conductive members, the electronic component and one surface of the relay plate are connected to each other by a first bonding wire, the one surface of the relay plate and one surface of the terminal member are connected to each other by a second bonding wire, the relay plate has a shape of a rectangular sheet, if a direction parallel to a long side of the relay plate is defined as a long side direction, each bonding wire is connected to a one-side portion or an other-side portion in the long side direction of the relay plate, the mold resin is formed in an external shape that is a shape of a rectangular sheet having a sheet surface parallel to the same plane, and the relay plate is disposed such that one long side thereof extends along a side surface of the mold resin.

2. The semiconductor module according to claim 1, wherein the electronic component is disposed on one side in the long side direction relative to the relay plate, the terminal member is disposed on

another side in the long side direction relative to the relay plate, and the first bonding wire is connected to the one-side portion in the long side direction of the relay plate.

3. The semiconductor module according to claim 2, wherein the first bonding wire connecting the electronic component and the one surface of the relay plate to each other, and the second bonding wire connecting the one surface of the relay plate and the one surface of the terminal member to each other, constitute a single bonding wire.

4. The semiconductor module according to claim 1, wherein the base plate, the relay plate, and the terminal member are made from a same material.

5. The semiconductor module according to claim 1, wherein each of the base plate, the relay plate, and the terminal member includes a terminal portion or an extending-out portion exposed to outside from a side surface of the mold resin, and directions in which the terminal portion and the extending-out portion are exposed are directions parallel to the same plane.

6. The semiconductor module according to claim 5, wherein a width of the extending-out portion of the relay plate is shorter than a length of the long side of the relay plate, and the extending-out portion of the relay plate is provided on a side, on which there is one long side of the relay plate, so as to be located at the one-side portion or the other-side portion in the long side direction of the relay plate.

7. The semiconductor module according to claim 3, wherein each of the base plate, the relay plate, and the terminal member includes a terminal portion or an extending-out portion exposed to outside from a side surface of the mold resin, directions in which the terminal portion and the extending-out portion are exposed are directions parallel to the same plane, a width of the extending-out portion of the relay plate is shorter than a length of the long side of the relay plate, and the extending-out portion of the relay plate is provided on a side, on which there is one long side of the relay plate, so as to be located at the one-side portion or the other-side portion in the long side direction of the relay plate.

8. The semiconductor module according to claim 7, wherein the base plate, the relay plate, and the terminal member are made from a same material.

9. Means for preventing wire sag and wire sweep in a semiconductor module, comprising: a base plate, a relay plate, and a terminal member being electrically conductive members and arranged as a same plane with gaps between the electrically conductive members; a first bonding wire connecting the electronic component and one surface of the relay plate, and a second bonding wire connecting the terminal member and the one surface of the relay plate; and a mold resin sealing the base plate, the relay plate, the terminal member, the electronic component, the first bonding wire, and the second bonding wire, wherein the relay plate has a shape of a rectangular sheet, if a direction parallel to a long side of the relay plate is defined as a long side direction, each bonding wire is connected to a one-side portion or an other-side portion in the long side direction of the relay plate, the mold resin is formed in an external shape that is a shape of a rectangular sheet having a sheet surface parallel to the same plane, and the relay plate is disposed such that one long side thereof extends along a side surface of the mold resin.
